**Clinical Study Summary Report (CSSR) Document Status:** Draft | **Version:** 1.0 | **Date:** 03-Apr-2026

---

## 1. Administrative & Study Identification

**Full Study Title:** A Phase 2, Open-Label, Single-Arm, Sequential-Panel Study to Evaluate the Pharmacokinetics, Safety, and Tolerability of Posaconazole (POS, MK-5592) Intravenous and Powder for Oral Suspension Formulations in Pediatric Participants From Birth to Less Than 2 Years of Age With Possible, Probable, or Proven Invasive Fungal Infection **Short Title/Acronym:** Posaconazole (MK-5592) Intravenous and Oral in Children (<2 Years) With Invasive Fungal Infection (MK-5592-127) **Protocol Number:** MK-5592-127 **NCT Number:** NCT04665037 **Posting ID:** 675145905589660818 **Sponsor/Company Name:** Merck **Investigational Product:** Posaconazole (POS, MK-5592) Intravenous (IV) and powder for oral suspension (PFS) **Phase of Development:** PHASE2 **Trial Status:** RECRUITING **Disease Areas:** Invasive Fungal Infection **Medical Monitor:** Merck Clinical Operations Lead

---

## 2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate and estimate the pharmacokinetics (PK) of posaconazole (POS, MK-5592) intravenous (IV) and powder for oral suspension (PFS) formulations in pediatric participants <2 years of age with invasive fungal infection (IFI). **Secondary Objectives:** To evaluate the safety and tolerability of single and multiple doses of POS IV and POS PFS formulations in neonates and infants.

**2.2 Background & Rationale** To address the clinical gap in proper dosing and exposure profiling for posaconazole in the highly vulnerable pediatric population ranging from birth to <2 years of age. Understanding the PK properties (such as $Cmax$ and $AUC$) is essential for ensuring appropriate therapeutic concentrations and minimizing toxicity when treating severe, life-threatening invasive fungal infections.

---

## 3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Single-arm (Sequential-Panel) **Blinding:** Open-label **Randomization:** N/A

**3.2 Planned Enrollment & Duration Targeted Enrollment ($N$):** 40 participants **Panel Distribution:** * Panel A: $\ge 8$ participants evaluating POS IV * Panel B: $\ge 14$ participants evaluating both POS IV and POS PFS (including $\ge 6$ who are <3 months of age and $\ge 5$ who transition to the PFS formulation of POS)

**Number of Sites:** ~15 global locations **Estimated Study Start:** 22-Feb-2022 **Estimated Primary Completion:** 16-Dec-2026

---

## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Age from 1 day (birth) to <2 years. 2. Panel A: Is undergoing treatment for possible, probable, or proven IFI known or suspected to be caused by fungal pathogens against which POS has demonstrated activity (which can include candidiasis). 3. Panel B: Has an investigator-assessed diagnosis of possible, probable, or proven IFI known or suspected to be caused by fungal pathogens against which POS has demonstrated activity (and cannot include candidiasis). 4. Has clinical symptoms consistent with an acute episode of IFI and has a central line (e.g., central venous catheter, peripherally-inserted central catheter) in place or planned to be in place before beginning IV study intervention. 5. Has a body weight of $\ge 500$ g. 6. The participant (or legally acceptable representative) has provided documented informed consent for the study.

**4.2 Exclusion Criteria** 1. Has received POS within 30 days before Day 1. 2. Has cystic fibrosis, pulmonary sarcoidosis, aspergilloma, or allergic bronchopulmonary aspergillosis. 3. Has a known hereditary problem of galactose intolerance, Lapp lactase deficiency, or glucose-galactose malabsorption (Panel B). 4. Has known or suspected active COVID-19 infection. 5. Has a known hypersensitivity or other serious adverse reaction to any azole antifungal therapy, or to any other ingredient of the study intervention used. 6. Has any known history of torsade de pointes, unstable cardiac arrhythmia or proarrhythmic conditions, a history of recent myocardial infarction, congenital or acquired QT interval prolongation, or cardiomyopathy in the context of cardiac failure within 90 days of first dose of study intervention. 7. Has received any listed prohibited medications within the specified timeframes before the start of study intervention. 8. Has enrolled previously in the current study and been discontinued. 9. Has QTc prolongation at screening >500 msec. 10. Has significant liver dysfunction. 11. Is hemodynamically unstable, exhibits hemodynamic compromise, or is not expected to survive at least 5 days. 12. Unsuitable for intensive blood sampling necessary for therapeutic drug monitoring and PK assessments.

---

## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Panel B: Posaconazole IV administered twice daily on Day 1, then once daily from Day 2 to a maximum of 84 days. Participants clinically able will transition to POS PFS on Day 8, administered once daily up to a maximum of 84 days. Panel A involves single-dose POS IV evaluation. **Route of Administration:** Intravenous (IV) infusion and Oral (Powder for oral suspension - PFS). **Duration of Treatment:** Up to 84 days of active treatment.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard-of-care antimicrobial therapy and baseline treatments for underlying morbidities. **Prohibited:** Concomitant administration of strong liver enzyme inducers/inhibitors (e.g., carbamazepine, phenobarbital, rifampin) that could confound pharmacokinetic analyses.

---

## 6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures |
| :--- | :--- | :--- |
| **Screening** | Day -14 to 0 | Informed Consent, Medical History, Baseline Safety Labs, IFI confirmation |
| **Baseline** | Day 1 | First Dose IV, Intensive PK Sampling, Safety Audit |
| **Interim** | Days 2 to 84 | Ongoing Safety Labs, PK Sampling, Transition to Oral PFS (Day 8 for Panel B) |
| **End of Study** | Follow-up | Final Vital Signs, Exit Interview, IP Accountability, Final Safety Evaluation |

---

## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoints** 1. Average concentration ($Cavg$) of single-dose IV POS (Panel A) based on population PK analysis. 2. Maximum concentration ($Cmax$) of single-dose IV POS (Panel A) based on population PK analysis. 3. Time to maximum concentration ($Tmax$) of single-dose IV POS (Panel A) based on population PK analysis.

**7.2 Secondary & Exploratory Endpoints** 1. $Cavg$ of IV POS in neonates and infants <2 years of age compared to adults and older pediatric populations (Panel B). 2. Percentage of participants with an $\ge 1$ adverse event (AE) (Panels A and B). 3. Percentage of participants who discontinued study therapy due to an AE (Panels A and B). 4. Overall clinical response and mortality mapping to antifungal exposure.

---

## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous collection via hospital/HCP monitoring, central lab abnormalities, and caregiver reporting. **Serious Adverse Events (SAEs):** Standard reporting timeline, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External, independent data safety monitoring board to evaluate pediatric PK data and toxicity limits.

---

## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Pharmacokinetic (PK) Set for concentration evaluation; Safety Set for all enrolled patients who receive at least one dose. **Sample Size Justification:** Targeted enrollment of $N=40$ is deemed adequate for pediatric population PK modeling to define drug clearance and volume of distribution in the under-2-years cohort. **Handling of Missing Data:** Standard pharmacokinetic non-compartmental analysis (NCA) imputation protocols; safety events utilizing Last Observation Carried Forward (LOCF) where required.

---

## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring visits initially due to complex PK blood draws; subsequent remote monitoring. **Audit Trail:** Robust Electronic Data Capture (EDC) with strict eCRF locking and query resolution tracking for specialized PK sampling times.

---

## 11. Terminology & Disease Ontology

**Preferred UMLS Name:** Invasive Fungal Infection **Disease Ontology Details:** * **MeSH Code**: D000072742 * **HPO Code**: HP:0020101 * **SNOMED ID**: 12391000132109 **Drug Terminology Details:** * **Reference Drug Name:** tolnaftate * **ATC Code:** D01AE18

---

## 12. Clinical Framework (Invasive Fungal Infection Specific)

**Primary Condition:** Invasive Fungal Infection (IFI) **Diagnosis Threshold:** Diagnosis of possible, probable, or proven IFI known or suspected to be caused by susceptible fungal pathogens **Medical Methodology:** Sequential antifungal management (IV to Oral step-down) **Optimization Target:** Target optimal $Cavg$ and steady-state exposure

---

## 13. Study Procedures & Methodology

**Management Pathway:** Intensive inpatient PK evaluation transitioning to outpatient oral suspension therapy **Education Protocol (HCP):** Specialized training on weight-based infant dosing, pediatric IV administration, and strict timed PK sampling **Education Protocol (Patient):** Caregiver education on the preparation and administration of Posaconazole powder for oral suspension (PFS) **Quality Audit Mechanism:** Central laboratory PK reconciliation and caregiver medication administration diary tracking

---

## 14. Observation & Follow-Up Schedule

**Total Duration:** ~12-16 weeks including treatment and safety follow-up **Onsite Visit Cadence:** Daily during the inpatient IV phase; periodic monitoring post-hospital discharge **Remote Monitoring Frequency:** Routine safety calls post-discharge during the PFS administration phase **Checklist Review Interval:** Daily diary review during the oral administration phase

---

## 15. Sign-off & Approval

| Role | Name | Signature | Date |
| :--- | :--- | :--- | :--- |
| **Principal Investigator** | [Name] | ____ | **03-Apr-2026** |
| **Clinical Operations Lead** | **[Name]** | ___ | ***03-Apr-2026*** |
| ***Lead Statistician*** | ***[Name]*** | _____ | 03-Apr-2026 |